

Cassandra

Cassandra is a NoSQL distributed database management system written in Java. It can handle large amounts of data across multiple data centres and in the cloud. Its ability to handle high volumes makes it particularly beneficial for major corporations such as Apple, Facebook, Instagram, Uber, Twitter, Cisco, Rackspace, eBay, and Netflix. It is the most trusted database for scalability and high availability with no single point of failure.

How to choose the right open source database for your needs

There are numerous database management systems available in the market. Choosing one can be very challenging for users. You have an array of options available in relational (MySQL, PostgreSQL, Oracle DB, etc) and non-relational (MongoDB, Apache HBase, Cassandra, etc) databases. But not one of them fits all kinds of project requirements. Each has some advantages as well as a few disadvantages.

Asking a few important questions related to your project can help identify the right database.

- How much data will be stored when the application is mature?
- How many users will handle data simultaneously at peak load?
- How much scalability, data consistency, latency, throughput and availability does your application need?
- What is the budget to cover licences and support contracts?
- How often will the database schemas change?
- What are the preferred programming languages?
- What is the geographic distribution of your user population?

Is open source the future of database management?

In the last few years, many companies have been turning away from proprietary software and adopting open source software. As a result, open source databases have become very popular.

These databases now support a range of modern applications, from the most popular mobile apps to leading e-commerce platforms. Fast-growing companies and large enterprises prefer open source databases due to their low cost, freedom from traditional licence models, flexibility, community-backed development and support, and large ecosystems of tools and extensions.

It is no surprise that if organisations want to use open source databases, they will also want to use them in the cloud, and companies such as Amazon are making that happen. AWS offers MySQL, PostgreSQL, MongoDB, and Redis, among others. The Microsoft Azure cloud supports PostgreSQL, MySQL, and others. Public cloud suppliers like Google, DigitalOcean, Rackspace, and others support a range of open source databases.

According to a 2019 open source data management software survey

conducted by database experts Percona, 89 per cent of developers are using at least one open source database. Gartner predicts that by 2022, 70 per cent of applications will feature some sort of open source database. Incorporating an open source database into your development workflow is becoming rather important. Soon, it will be non-optional.

According to a CNBC report, MongoDB's market cap swelled past US\$ 32 billion in September 2021, making it the most valuable open source company on record. Companies such as MongoDB and Databricks, which have developed market-leading products that work across major cloud platforms built by Amazon, Microsoft and Google, are thriving as customers invest in moving their data and applications from traditional data centres to the cloud.

Just like the emergence of open source software has transformed the software industry, open source data management systems have transformed the database solutions industry. Open source DBMSs are popular because of their flexibility and cost-effectiveness, and are making organisations take a relook at the database solutions they use. **END** 

References

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